

Benchmarking Dispatching Algorithms Using Real Scenarios in Elevator Monte Carlo Simulation

Gastón Amengual
Facultad Regional Rosario
Universidad Tecnológica Nacional
Rosario, Argentina
gamengual@frro.utn.edu.ar

Ezequiel L. Castaño
Facultad Regional Rosario
Universidad Tecnológica Nacional
Rosario, Argentina
ecastano@frro.utn.edu.ar

Abstract—Elevators are present in most universities, offices, and residential buildings, and people rely heavily on them for their transportation within the building. Therefore, when the performance of elevators is not good enough, they are usually the main source of complaints. This performance depends mainly on their internal logic or dispatching algorithm. It is then important to know which dispatching algorithm is the best and whether the elevator will perform similarly in both low and high-demand situations. The present work presents a benchmark on how different dispatching algorithms impact the time metrics for the users. To produce this benchmark, a Monte Carlo simulation of a single elevator was modeled, analyzing the user waiting time in the queue, in the elevator, and in the system.

Index Terms—Elevator, Dispatching algorithm, Monte Carlo Simulation

I. INTRODUCTION

Elevators are the most common means of transport inside buildings like universities, offices, and residential buildings. Although elevators have evolved significantly since their inception, the core functions remain the same, and one of the key issues concerning their usage is still the time the users spend in the queue and in the elevator.

Even though elevators may differ in several aspects like the technology used, the safety mechanisms, and their physical dimensions, the most important feature that determines the waiting time for the users is the dispatching algorithm used, i.e. the logic the elevator uses to decide the next call to be served. It becomes useful analyzing different high-demand contexts for elevators, comparing several dispatching algorithms, and evaluating their impact on the aforementioned user times. Then, the following questions arise:

- 1) Which is on average the best performing dispatching algorithm?
- 2) Is the level (low/high) of the demand relevant for determining the best dispatching algorithm?

The first question refers to finding an optimal solution, and the second evaluates whether the best solution still performs well under demand level variations. The present work presents a discrete-event Monte Carlo simulation of a single elevator, applying different dispatching algorithms and measuring the time in the queue and in the elevator time.

In the methodology section, the model assumptions will be described, a brief explanation of the simulation model will be presented, the dispatching algorithms will be introduced,

explaining the different simulation scenarios in which they will be analyzed. The chosen metrics will be explained, and the results will be shown. Finally, conclusions regarding the experiments and the results will be drawn.

II. LITERATURE REVIEW

Extensive research has been done on elevators focusing on other specific topics. Safety and emergency measures have been approached by [1]. Hardware, including integrated circuits, boards, and multiplexers has been studied by [2] and [3]. Physical models and their implications were approached by [4] and [5]. Energy consumption was studied by [6]. Fuzzy logic controller models were proposed by [7] and [8]. Finally, tall buildings were studied by [9] and [10].

About works focused on analyzing scenarios, performance measures, and elevator dispatching algorithms, [11] provides a simulation model program written in Simula for the morning rush hour in an office building, [12] provides a simulation model implemented in DOS, [13] presents an elevator system in Arena simulation software, modeling an office building, and provides a dispatching algorithm and three variations of the former, [14] compares a Monte Carlo elevator simulation in MATLAB to the analytical formulae, [15] presents a discrete event simulation queuing theory model for an elevator traffic system built using the SimEvents blocks within Simulink (MATLAB), [16] deals with sectoring, [17] models the simulation of the elevator of a particular university, using Anylogic Simulation Modeling Software, and [18] considers a simulation model based on a Hospital located in the north of Portugal.

It can be seen that elevators have been simulated in several works and also using different tools. However, no prior work has been done to benchmark different dispatching algorithms on different real scenarios using a discrete-event simulation, which is the main contribution of this work.

III. METHODOLOGY

To answer the triggering questions, it is necessary to state the assumptions and describe the simulation model, the dispatching algorithms to be used, the relevant scenarios, and the performance metrics.

A. Model Assumptions

A simulation model can only represent the real world with some simplifications, in this case, the following assumptions were made to reduce the inherent complexity of the real system:

- 1) There is one elevator.
- 2) The building has less than 10 floors.
- 3) The elevator opens and closes automatically.
- 4) The elevator maximum capacity is given by the number of users.
- 5) The elevator velocity is assumed constant, i.e. it takes the elevator a fixed amount of time to go to the immediate next floor.
- 6) On each floor there is only one elevator button that does not distinguish between up and down.

Concerning the user, the following assumptions are made:

- When a user arrives at a certain floor, it joins the queue of people on that floor.
- The time between user arrivals depends on the floor, and it follows a Gamma distribution.
- A user leaves the queue if certain waiting tolerance is exceeded.
- A user never changes its destination throughout the simulation.
- A user never enters the elevator if there is not enough space.
- A user does not manually stop the elevator.
- A user can stop the doors from closing and enter the elevator.

B. Model

The elevator was modeled using a discrete-event simulation [19] that consists of the following events:

- **User arrival:** it controls the user behavior when they arrive, either they take the elevator if it is on the floor, or otherwise they press the button and get into the queue.
- **Process calls:** it calls the dispatching algorithm to compute the elevator next floor, and calls the two following events depending on the current floor.
- **Move elevator one floor:** it moves the elevator one floor up or down, in direction to the next destination floor.
- **Load and unload in floor:** it controls the loading and unloading of users on a destination floor, the users inside leave, and the users outside enter if there is enough space.
- **Enter elevator:** it lets users enter the elevator.

C. Dispatching Algorithms

A dispatching algorithm, also known as elevator group control, is the algorithm that manages all the elevator calls, deciding to which floor the elevator must go next [20].

The term *outside call* will be used to refer to calls made from a floor queue, while *inside call* will be used to refer to calls made from inside the elevator. For all the following algorithms, if there are no calls, the elevator remains idle on the last floor it was called from.

1) **FIFO:** The First-In, First-Out algorithm satisfies the elevator calls by time of arrival, without distinguishing between outside and inside calls.

2) **IFOS:** The Inside-First, Outside-Second algorithm satisfies the elevator calls by time of arrival, taking into consideration that: (1) If there are inside calls, the elevator satisfies the first inside call. (2) Otherwise, the elevator satisfies the first outside call.

3) **LOOK:** The LOOK algorithm works as follows: (1) The elevator moves up satisfying all calls. If there are no more calls for floors above, it goes down. (2) The elevator moves down satisfying all calls. If there are no more calls for floors below, it goes up. This algorithm does not distinguish between outside and inside calls.

4) **LOOK-M:** The LOOK-Modified is a modified version of LOOK, and it works as follows: (1) The elevator moves up satisfying only inside calls. Once all upper inside calls were satisfied, it satisfies all upper outside calls. If there are no more calls for floors above, it goes down. (2) The elevator moves down satisfying both inside and outside calls. If there are no more calls for floors below, and there are calls to or from upper calls, it goes up.

Both FIFO and LOOK are introduced and explained in detail in [21], the IFOS and LOOK-M are methods proposed by the authors.

D. Scenarios

To simulate real-life situations, different high-demand and low-demand scenarios were considered. The demand is defined as the number of users calling the elevator, both from outside and from inside. Then, to test the dispatching algorithms, the following scenarios were built:

- 1) **Morning Rush:** (high-demand) there is a high demand from the ground floor to go to all other floors, from which there is no demand. This scenario represents the early hours for most universities and offices, where the students or workers enter the building immediately after the building has opened and go up to all other floors.
- 2) **Lunchtime:** (high-demand) there is a high demand from all floors to go to the restaurant floor (considered as the first floor) and to the ground floor (for users that have lunch outside). Moreover, there is a high demand from the restaurant and ground floor of the users that return to work.
- 3) **Evening Rush:** (high-demand) there is a high demand from all floors to go to the ground floor and no demand from the ground floor. This is usual in universities and offices, where the students or workers leave the building soon before it closes.
- 4) **Residential Building:** (low-demand) there is low demand from all floors. This scenario is usual in both residential buildings and universities and offices outside the rush hours and lunchtimes. Inter-floor traffic is not considered, most traffic is from different floors to the ground floor, and from the ground floor to different floors.

For all cases, the destination distribution was considered equal except for contiguous floors to the current floor, assuming that for such cases most people would simply take the stairs.

To better reflect a typical business day, a combination of the above-mentioned scenarios will be used, according to the following schedule:

- 1) 0.5 hr. Morning Rush
- 2) 3 hr. Residential Building
- 3) 1 hr. Lunchtime
- 4) 3 hr. Residential Building
- 5) 0.5 hr. Evening Rush

E. Performance Metrics

Once the simulation is run, several performance metrics were calculated for comparison purposes. The variable of interest is related to the time spent by users using the elevator. This is the most recurrent variable in other works such as [11], [12], [14], [18]. Specifically, three times were considered:

- 1) **Queue time:** the time that a user spends waiting in a floor queue until he or she enters the elevator. This is probably the most important time concerning user satisfaction, as discussed in [22].
- 2) **Elevator time:** the time it takes for the user to go to the destination floor once in the elevator.
- 3) **System time:** the time since the user enters the system until it leaves the system, i.e. the sum of the Queue and Elevator times.

IV. EXPERIMENT DESIGN

The simulation model described was implemented in Python. Each possible combination of dispatching algorithm and scenario was run 200 times. For each run, the queue, elevator, and system times were recorded.

For the setting of the simulation, the building consisted of the ground floor and 5 floors, the elevator capacity was 6 people, the elevator took 3 seconds to close the doors, 5 seconds to go from one floor to the contiguous floor, and it took 1 second for a person to enter the elevator.

V. RESULTS

The mean values of the results for Queue, Elevator, and System time are shown in Table I, Table II, and Table III, respectively, and a summary of the same values can be visualized in Fig. 1. The confidence intervals were calculated but omitted due to their extremely small width.

In all three high-demand scenarios (Morning Rush, Evening Rush, Lunchtime), both LOOK (606.99, 112.49, 1075.82) and LOOK-M (607.11, 112.50, 1075.52) algorithms achieved significantly shorter system times than FIFO (647.97, 646.50, 1194.99) and IFOS (633.64, 647.32, 1109.35). For all scenarios, there was no significant difference between LOOK and LOOK-M in queue (19.60; 20.35, 572.36; 572.44, 58.84; 58.87, 1031.54; 1031.30, 773.51; 773.50) and system times (45.68; 45.52, 606.99; 607.11, 112.49; 112.50, 1075.82; 1075.52, 817.80; 817.69), and in elevator time there was a difference only in Residential Building (26.08; 25.18), where LOOK-M outperformed LOOK. Overall, the FIFO dispatching algorithm had the longest elevator time, with its highest mean value being 86.41 seconds. IFOS, on the other hand, achieved a much shorter elevator time, especially in the Evening Rush scenario (23.85), where it outperformed the other algorithms by a significant margin (86.41, 53.62, 53.62). The variance between dispatching algorithms when considering the Residential Building scenario is concentrated in the elevator time, as in queue and system times all algorithms performed similarly (23.32, 34.43, 19.60, 20.35). Even though all dispatching algorithms performed similarly regarding the queue time in Morning Rush (597.67, 590.65, 572.36, 572.44), in Evening Rush LOOK (58.84) and LOOK-M (58.87) were up to 9.5 times faster than FIFO (559.97) and 10.5 times faster than IFOS (623.47). Concerning the Business Day scenario, both LOOK and LOOK-M performed similarly for all metrics (773.51; 773.50, 44.30; 44.26, 817.80; 817.69). The FIFO was consistently the slowest (952.94, 57.95, 1010.89), and IFOS was the fastest regarding elevator time (40.21). Overall, both LOOK and LOOK-M outperformed FIFO and IFOS in system time by a significant margin, specially in Evening Rush, when LOOK and LOOK-M were approximately 6 times faster time faster than the rest with a value of (112.49 and 112.50 against 646.50 and 647.32).

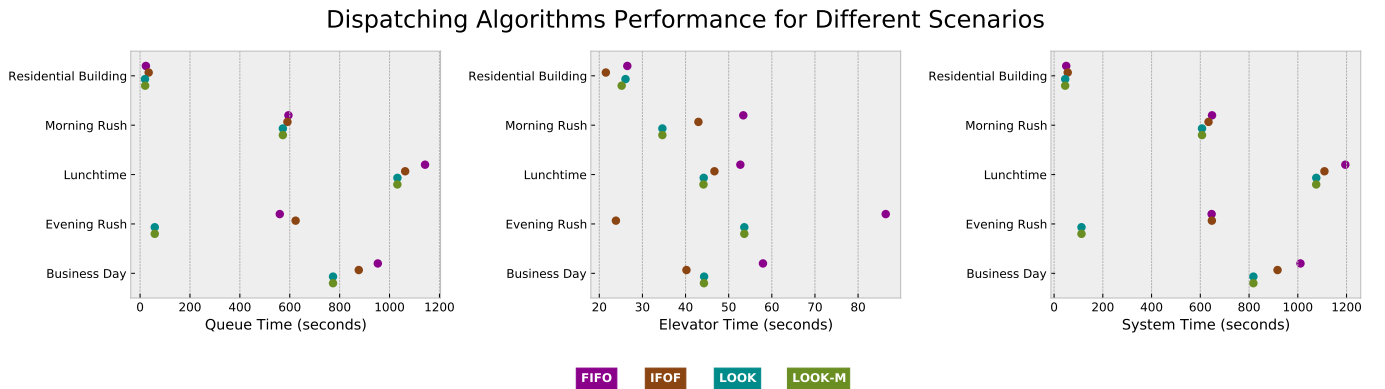


Fig. 1. Performance metrics for different dispatching algorithms and scenarios

<i>Queue Time</i>	FIFO	IFOS	LOOK	LOOK-M
Residential Building	23.32	34.43	19.60	20.35
Morning Rush	597.67	590.65	572.36	572.44
Evening Rush	559.97	623.47	58.84	58.87
Lunchtime	1142.27	1062.63	1031.54	1031.30
Business Day	952.94	876.71	773.51	773.50

TABLE I
QUEUE TIME RESULTS

<i>Elevator Time</i>	FIFO	IFOS	LOOK	LOOK-M
Residential Building	26.50	21.52	26.08	25.18
Morning Rush	53.38	42.98	34.63	34.63
Evening Rush	86.41	23.85	53.62	53.62
Lunchtime	52.70	46.70	44.21	44.16
Business Day	57.95	40.21	44.30	44.26

TABLE II
ELEVATOR TIME RESULTS

<i>System Time</i>	FIFO	IFOS	LOOK	LOOK-M
Residential Building	49.81	55.95	45.68	45.52
Morning Rush	647.97	633.64	606.99	607.11
Evening Rush	646.50	647.32	112.49	112.50
Lunchtime	1194.99	1109.35	1075.82	1075.52
Business Day	1010.89	916.86	817.80	817.69

TABLE III
SYSTEM TIME RESULTS

VI. CONCLUSIONS

Benchmarking dispatching algorithms for elevators is one of the most straightforward ways to assess the best elevator implementation for a given scenario. In this work, several scenarios and dispatching algorithms were simulated to approximate and analyze a wide range of real-world-like situations.

There is no clear best-performing dispatching algorithm, and the selection and adoption of one algorithm depend on the scenario being considered. It is concluded, also, that the demand plays a crucial role when determining the best dispatching algorithm. Overall, the following conclusions can be drawn. This work shows that considering whether the call comes from outside or inside plays a pivotal role when determining the results. The FIFO algorithm is the worth algorithm possible for most scenarios. IFOS is the best general choice when elevator time is critical for the business, but at the expense of queue time. The LOOK and LOOK-M dispatching algorithms are the best choices for all scenarios considering the overall system time. In the case of low-demand buildings, LOOK-M should be preferred, while, for other cases, there is no significant difference between the two. Finally, if the queue waiting time is critical for the business, then LOOK or LOOK-M should be selected.

VII. FUTURE WORK

This work presented a baseline for future comparisons. It allows identifying which a good starting point is regarding performance metrics. In future work, more complex and possible non-deterministic dispatching algorithms may be used for benchmarking, involving, for example, the use of reinforcement learning techniques, nature-inspired algorithms, and

fuzzy logic. Moreover, some assumptions can be relaxed, considering elevator acceleration, multiple elevators, and up and down buttons.

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